

A Practical Guide to Sigmoid function

TOPIC -- SIGMOID FUNCTION

-- 01 --

Gudermannian function $f(x) = \operatorname{gd} \blacksquare(x) = \int_0^x \operatorname{d} t \cosh \blacksquare t = 2 \arctan \blacksquare \left(\tanh \blacksquare(x^2) \right)$ $\{ \displaystyle f(x) = \operatorname{gd} \blacksquare(x) = \int_0^x \frac{dt}{\cosh t} = 2 \arctan \left(\tanh \left(\frac{x}{2} \right) \right) \}$

-- 02 --

$f(x) = \left\{ \frac{2}{1 + e^{-2mx}} - 1, |x| < 1 \operatorname{sgn} \blacksquare(x) |x| \geq 1 \right\} \tanh \blacksquare \left(m \sqrt{1 - x^2} \right), |x| < 1 \operatorname{sgn} \blacksquare(x) |x| \geq 1$ $\{ \displaystyle \begin{aligned} f(x) &= \begin{cases} \frac{2}{1 + e^{-2mx}} - 1, & |x| < 1 \\ \operatorname{sgn}(x) & |x| \geq 1 \end{cases} \\ &= \begin{cases} \tanh \left(m \sqrt{1 - x^2} \right), & |x| < 1 \\ \operatorname{sgn}(x) & |x| \geq 1 \end{cases} \end{aligned} \}$ using the hyperbolic tangent mentioned above. Here, m $\{ \displaystyle m \}$ is a free parameter encoding the slope at $x = 0$ $\{ \displaystyle x = 0 \}$, which must be greater than or equal to $\sqrt{3}$ because any smaller value will result in a function with multiple inflection points, which is therefore not a true sigmoid. This function is unusual because it actually attains the limiting values of -1 and 1 within a finite range, meaning that its value is constant at -1 for all $x \leq -1$ $\{ \displaystyle x \leq -1 \}$ and at 1 for all $x \geq 1$ $\{ \displaystyle x \geq 1 \}$. Nonetheless, it is smooth (infinitely differentiable, C^∞ $\{ \displaystyle C^\infty \}$) everywhere, including at $x = \pm 1$ $\{ \displaystyle x = \pm 1 \}$.

-- 03 --

Smoothstep function $f(x) = \left\{ \left(\int_0^1 (1 - u^2)^N \operatorname{d} u \right) - 1 \int_0^x (1 - u^2)^N \operatorname{d} u, |x| \leq 1 \operatorname{sgn} \blacksquare(x) |x| \geq 1 \right\} N \in \mathbb{Z} \geq 1$ $\{ \displaystyle f(x) = \begin{cases} \left(\int_0^1 (1 - u^2)^N \operatorname{d} u \right) - 1 \int_0^x (1 - u^2)^N \operatorname{d} u, & |x| \leq 1 \\ \operatorname{sgn}(x) & |x| \geq 1 \end{cases} \quad N \in \mathbb{Z} \geq 1 \}$

-- 04 --

Error function $f(x) = \operatorname{erf} \blacksquare(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} \operatorname{d} t$ $\{ \displaystyle f(x) = \operatorname{erf} \blacksquare(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} \operatorname{d} t \}$

-- 05 --

Further reading Mitchell, Tom M. (1997). Machine Learning. WCB McGraw–Hill. ISBN 978-0-07-042807-2.. (NB. In particular see "Chapter 4: Artificial Neural Networks" (in particular pp. 96–97) where Mitchell uses the word "logistic function" and the "sigmoid function" synonymously – this function he also calls the "squashing function" – and the sigmoid (aka logistic) function is used to compress the outputs of the "neurons" in multi-layer neural nets.) Humphrys, Mark. "Continuous output, the sigmoid function". Archived from the original on 2022-07-14. Retrieved 2022-07-14. (NB. Properties of the sigmoid, including how it can shift along axes and how its domain may be transformed.)

-- 06 --

Hyperbolic tangent (shifted and scaled version of the logistic function, above) $f(x) = \tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}}$

-- 07 --

Many natural processes, such as those of complex systems learning curves, exhibit a progression from small beginnings that accelerates and approaches a climax over time. When a specific mathematical model is lacking, a sigmoid function is often used. The van Genuchten–Gupta model is based on an inverted S-curve and applied to the response of crop yield to soil salinity. Examples of the application of the logistic S-curve to the response of crop yield (wheat) to both the soil salinity and depth to water table in the soil are shown in modeling crop response in agriculture. In artificial neural networks, non-smooth functions are sometimes used for efficiency; these are known as hard sigmoids. In audio signal processing, sigmoid functions are used as waveshaper transfer functions to emulate the sound of analog circuitry clipping. In Digital signal processing in general, sigmoid functions, due to their higher order of continuity, have much faster asymptotic rolloff in the frequency domain than a Heavyside step function, and therefore are useful to smooth discontinuities before sampling to reduce aliasing. This is, for example, used to generate square waves in many kinds of Digital synthesizer. In biochemistry and pharmacology, the Hill and Hill–Langmuir equations are sigmoid functions. In computer graphics and real-time rendering, sigmoid functions are used to blend colors or geometry between two values, producing smooth transitions without visible seams or discontinuities. Titration curves between strong acids and strong bases have a sigmoid shape due to the logarithmic nature of the pH scale. The logistic function can be calculated efficiently by utilizing type III Unums. A hierarchy of sigmoid growth models with increasing complexity (number of parameters) was built with

the primary goal to re-analyze kinetic data, the so-called N-t curves, from heterogeneous nucleation experiments, in electrochemistry. The hierarchy includes at present three models, with 1, 2, and 3 parameters, if not counting the maximal number of nuclei $N_{\max}$, respectively—a tanh² based model called $\alpha 21$ originally devised to describe diffusion-limited crystal growth (not aggregation!) in 2D, the Johnson–Mehl–Avrami–Kolmogorov (JMAK) model, and the Richards model. It was shown that for the concrete purpose, even the simplest model works, and thus it was implied that the experiments revisited are an example of two-step nucleation with the first step being the growth of the metastable phase in which the nuclei of the stable phase form.

-- 08 --

Properties In general, a sigmoid function is monotonic, and has a first derivative which is bell shaped. Conversely, the integral of any continuous, non-negative, bell-shaped function (with one local maximum and no local minimum, unless degenerate) will be sigmoidal. Thus the cumulative distribution functions for many common probability distributions are sigmoidal. One such example is the error function, which is related to the cumulative distribution function of a normal distribution; another is the arctan function, which is related to the cumulative distribution function of a Cauchy distribution. A sigmoid function is constrained by a pair of horizontal asymptotes as $x \rightarrow \pm \infty$ $\{\displaystyle x \rightarrow \pm \infty\}$. A sigmoid function is convex for values less than a particular point, and it is concave for values greater than that point: in many of the examples here, that point is 0.